

Neal Miller

ENGINEERING LEADER · AI-FIRST ENGINEERING · PRODUCT DELIVERY · ARCHITECTURAL TRANSFORMATION & MODERNIZATION

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Summary

Engineering leader with a proven track record building and scaling multi-team engineering organizations — growing from ~8-person teams to orgs of 30+ at Imagine Learning alone — across complex, high-stakes product environments. That includes founding Imagine Learning's first cross-product platform engineering team, run deliberately lean and AI-leveraged. Currently leading engineering for EdgeEX, Imagine Learning's market-leading courseware platform, across 4 teams, with a focus on modernization, cross-product infrastructure, and long-term technical strategy. A practitioner and organizational champion of agentic, AI-first development, having led an early end-to-end adoption of this model in a commercial engineering organization. Ph.D., Columbia University.

Core Competencies

Leadership & Strategy

Organizational leadership, Multi-team management, Technical strategy, Product alignment, Hiring & talent development, Vendor/contract management, Agile & PI planning, AI-first SDLC transformation, Org design, Portfolio-level prioritization, Change management

Technical Expertise

Service-oriented architecture (SOA), Event-driven systems, Distributed systems, Scalable web applications, System design, Data-intensive applications, Cloud-native modernization, Architectural governance

AI & Engineering Practice

Agentic development workflows, AI-augmented delivery, LLM tooling & integration, Engineering productivity, Claude Code / GitHub Copilot

Experience

Senior Software Engineering Manager

Remote

IMAGINE LEARNING

June 2025 - Present

Promoted to Senior Software Engineering Manager in June 2025 to lead the **Literacy Suite** (3 teams, ~20 engineers, led through a mix of direct reports and a team-level manager). Moved in May 2026 to lead engineering for **EdgeEX** — a step up in strategic visibility, as Imagine Learning's flagship courseware platform serving ~400K high school students and the strategic successor to its legacy Edgenuity platform — now across 4 teams of ~32 engineers and contractors, reporting directly to the CTO. Also founded Imagine Learning's first cross-product platform engineering team, **Learning Platform**, later handing off its leadership to another engineering manager.

Leadership & Delivery

- Lead ~32 engineers and contractors across 4 teams, with ~14 direct reports, including a distributed team based in the US and India.
- Drive cross-team architectural alignment, dependency management, and delivery execution across EdgeEX's engineering teams.
- Partner with Product, Design, and Curriculum leadership to define roadmap priorities and deliver high-impact, user-centered solutions.
- Grew talent within the org, including promoting a Literacy Suite engineer (early 2026) and onboarding 7 engineers through Imagine Learning's new India office expansion.

Technical & Organizational Impact

- Built Learning Platform as a deliberately lean, AI-leveraged team, establishing it as Imagine Learning's first adopter of an agentic, AI-first development model.
- Champion adoption of AI-assisted development tools (Claude Code, GitHub Copilot) — teams applying them saw user-story delivery increase ~107% versus Q4 2025 (comparable teams), improving productivity and code quality.
- Initiated a multi-year effort to modernize legacy monoliths within the Literacy Suite, handing it off upon transitioning to EdgeEX.

Software Engineering Manager

Remote

IMAGINE LEARNING

June 2023 - June 2025

Led the Literacy Data & Insights team, delivering real-time educator-facing reports within the Literacy Suite, which serves **1M+** students.

- Delivered the Growth Report for BTS 2024 — the most technically complex report in the Literacy Suite — spanning district, school, group, and student views on a new real-time data architecture.
- Managed a team of up to ~8 software engineers and SDETs (mix of FTEs and contractors).
- Ensured alignment through quarterly PI planning, managing cross-team dependencies and engineering commitments.
- Contributed to hiring decisions for both FTEs and contractors, and managed vendor relationships.

Software Engineering Manager

Remote

SMILEDIRECTCLUB

June 2022 - June 2023

Promoted to Software Engineering Manager to lead the Core Services team, building foundational internal services supporting user-facing products across SmileDirectClub's engineering organization, including contributing infrastructure supporting **SmileOS** — the company's AI-powered treatment planning platform serving 1.5M+ customers.

- Managed a blended team of ~6 engineers and contractors.
- Drove technical alignment and prioritization across product, business, and engineering stakeholders.
- Served as the primary liaison between Core Services and the broader organization, ensuring clarity of communication and strategic alignment.

Software Engineering Team Lead

Remote

SMILEDIRECTCLUB

August 2020 - June 2022

- **Partner Network**
 - Transformed team delivery practices — eliminating dysfunctional sprint ceremonies and transitioning from multi-hour nightly releases to on-demand deployments with minimal overhead.
 - Led development of the Partner Portal supporting SDC's Partner Network initiative.
- **Prescriptions**
 - Led the transition of the Prescriptions domain from a monolithic Django application to an event-driven, service-oriented architecture.
 - Delivered the Treatment Plan Service and shifted all functionality with zero downtime and minimal disruption.

Senior Lead Software Developer

Houston, TX

FRANCHICZAR

March 2019 - August 2020

Led technical direction for a 6-person development team, driving architecture, delivery, and hands-on development across multiple products.

- Played a central role in hiring and onboarding of engineering team members.
- Oversaw development of multiple products, including:
 - FranchiCzar OS – CRM for franchise-based sales
 - Fund.io – payment and subscription management platform
 - Rylos – learning management system (LMS)

Software Engineer Team Lead

Houston, TX

GENE BY GENE

November 2017 - March 2019

Senior Software Developer

Houston, TX

DURWELLA

February 2016 - November 2017

Lead Software Engineer

Houston, TX

RIGMINDER

June 2014 - February 2016

Learning Analytics Engineer

Houston, TX

REASONING MIND

July 2012 - June 2014

Developed algorithms for educational data mining within the Reasoning Mind adaptive learning platform, co-authoring peer-reviewed research on automated detection of student engagement and teacher behavior across a cohort of 70,000 students.

Education

Columbia University in the City of New York

New York, NY

PH.D. CHEMICAL PHYSICS (WITH DISTINCTION)

2012

- Dissertation: *Self-organization in systems of anisotropic particles*

The University of Texas at Austin

Austin, TX

B.S. CHEMISTRY, HONORS (WITH HIGHEST HONORS)

2007